

**Clinical Study Summary Report (CSSR)**  
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**1. Administrative & Study Identification**

**Full Study Title:** A Clinical Study of Nemtabrutinib in Japanese Participants With Hematological Malignancies [MK-1026-002]  
**Short Title/Acronym:** BELLWAVE-002 / MK-1026-002  
**Protocol Number:** MK-1026-002  
**NCT Number:** NCT05673460  
**Sponsor Name:** Merck Sharp & Dohme LLC  
**Investigational Product:** Nemtabrutinib (MK-1026)  
**Phase of Development:** Phase 1  
**Medical Monitor:** Merck Sharp & Dohme LLC Clinical Operations Lead

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**2. Study Synopsis & Rationale**

**2.1 Study Objective Primary Objective:** To evaluate the safety, tolerability, and pharmacokinetics (PK) of nemtabrutinib in Japanese participants with mature B-cell neoplasms. **Secondary Objectives:** To evaluate the preliminary efficacy, including Objective Response Rate (ORR), of nemtabrutinib in this patient population.

**2.2 Background & Rationale** To address the need for effective targeted therapies in patients with B-cell malignancies who have developed resistance or intolerance to prior treatments. Nemtabrutinib is a non-covalent, reversible, ATP-competitive Bruton's tyrosine kinase (BTK) inhibitor designed to maintain efficacy against both wild-type BTK and tumors bearing the C481 mutation, which commonly drives resistance to first-generation BTK inhibitors. This study establishes the safety and PK profile specifically in the Japanese population.

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**3. Study Design & Methodology**

**3.1 Design Framework Study Type:** Interventional  
**Control Type:** Single-arm  
**Blinding:** Open-label  
**Randomization:** N/A

**3.2 Planned Enrollment & Duration Target \$N\$:** 7 evaluable Japanese participants  
**Number of Sites:** Multiple sites in Japan  
**Estimated Study Start:** 14-Dec-2022 **Estimated Primary Completion:** Completed

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## 4. Subject Selection (Eligibility Criteria)

**4.1 Inclusion Criteria** 1. Age  $\geq 18$  years old; must be Japanese. 2. Histologically confirmed B-cell malignancy: Chronic lymphocytic leukemia (CLL), Small lymphocytic lymphoma (SLL), Waldenström's macroglobulinemia (WM), Lymphoplasmacytic lymphoma (LPL), or other mature B-cell neoplasm. 3. Failed or intolerant to at least 2 prior regimens given in combination or sequentially, OR have received 1 prior BTK-containing regimen when a BTK inhibitor is approved as first-line therapy. 4. Ability to swallow and retain oral medication.

**4.2 Exclusion Criteria** 1. Active Hepatitis B virus (HBV) or Hepatitis C virus (HCV) infection at study entry. 2. Known history of human immunodeficiency virus (HIV) infection. 3. History of a second malignancy, unless potentially curative treatment has been completed with no evidence of malignancy for 3 years. 4. Clinically significant gastrointestinal abnormalities that might alter absorption (e.g., gastric bypass surgery, gastrectomy) or underlying severe bleeding disorders. 5. History or concurrent condition of pneumonitis/interstitial lung disease.

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## 5. Intervention & Treatment Plan

**5.1 Investigational Regimen Dosage/Frequency:** Nemtabrutinib administered orally at designated dose levels. **Route of Administration:** Oral (Tablet). **Duration of Treatment:** Administered continuously in 28-day cycles until disease progression, unacceptable toxicity, or withdrawal of consent.

**5.2 Concomitant & Prohibited Therapy Permitted:** Standard supportive care for hematological malignancies. **Prohibited:** Concomitant anti-cancer treatments and other investigational drugs during the study duration. Washout periods apply for prior therapies.

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## 6. Study Timeline & Visit Schedule

| Visit ID           | Window (Days) | Key Activities/Procedures                                                                                       |
|--------------------|---------------|-----------------------------------------------------------------------------------------------------------------|
| Screening          | Day -28 to 0  | Informed Consent, Medical History, Baseline Labs, Disease Assessment                                            |
| Baseline (Cycle 1) | Day 1         | First Dose, Intensive PK/Safety monitoring, Vital Signs                                                         |
| Interim Visits     | Cycle 1-2     | Ongoing PK sampling (C <sub>max</sub> , C <sub>min</sub> , AUC, T <sub>max</sub> ), DLT evaluation, Safety Labs |
| Follow-Up          | Every Cycle   | Efficacy Assessments (ORR), Safety Monitoring, IP Accountability                                                |
| End of Study       | +30 Days      | Final Safety Check, Follow-up for survival and disease progression                                              |

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## 7. Outcome Measures (Endpoints)

**7.1 Primary Endpoint Definition:** 1. Number of participants experiencing Dose Limiting Toxicities (DLTs). 2. Number of participants with any Adverse Events (AEs) and Serious Adverse Events (SAEs). 3. Pharmacokinetic profile (C<sub>max</sub>, T<sub>max</sub>, AUC, C<sub>min</sub> of plasma nemtabrutinib). **Measurement Method:** Toxicities graded using NCI-CTCAE v5.0, except hematologic toxicities in CLL/SLL which are assessed via iwCLL criteria. PK analyzed via centralized blood sampling.

**7.2 Secondary & Exploratory Endpoints Objective Response Rate (ORR):** Proportion of participants achieving at least a partial response (PR) per disease-specific evaluation criteria as assessed by the investigator.

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## 8. Safety & Adverse Event Reporting

**Adverse Event (AE) Monitoring:** Continuous monitoring of untoward medical occurrences temporally associated with the use of study intervention. **Serious Adverse Events (SAEs):** Reported within 24 hours to the sponsor and appropriate regulatory bodies. **Data Monitoring Committee (DMC):** Sponsor-led internal safety review committee for dose escalation and DLT assessment.

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## 9. Statistical Analysis Plan (SAP) Summary

**Analysis Populations:** Safety Set (all participants receiving at least 1 dose of nemtabrutinib); PK Set (participants with sufficient concentration-time data). **Sample Size Justification:**  $N=7$  was chosen based on standard Phase 1 safety/PK principles to identify DLTs and confirm the PK profile in the Japanese population prior to broader cohort expansion. **Handling of Missing Data:** Missing data handles inherently without imputation for PK analysis; LOCF or standard oncological data handling applied for efficacy where applicable.

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## 10. Quality Assurance & Regulatory Compliance

**GCP Compliance:** This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Standard sponsor-directed onsite and remote monitoring to assure data integrity, AE reporting, and protocol adherence. **Audit Trail:** Maintained through secure electronic Data Capture (eDC) systems with query resolution and eCRF locking.

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11. Study Identifiers & Governance

**Primary ID:** MK-1026-002 **Registry Number:** NCT05673460 **Sponsor/Lead Organization:** Merck Sharp & Dohme LLC **Study Title:** BELLWAVE-002 **Official Regulatory Title:** A Clinical Study of Nemtabrutinib in Japanese Participants With Hematological Malignancies (MK-1026-002)

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12. Clinical Framework (Nemtabrutinib Specific)

**Primary Condition:** Hematologic Malignancies (Mature B-cell Neoplasms) **Diagnosis Threshold:** Histologically confirmed CLL, SLL, WM, LPL, or other B-cell neoplasm **Medical Methodology:** Non-covalent Tyrosine Kinase Inhibition (Targeted BTK inhibitor) **Optimization Target:** Disease response (ORR) and tolerable safety profile

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13. Study Procedures & Methodology

**Management Pathway:** Oral targeted anti-cancer clinical pathway **Education Protocol (HCP):** Site initiation visits and clinical protocol training on AE/DLT reporting standards **Education Protocol (Patient):** Informed consent detailing dosing schedule, fasting requirements, and side-effect reporting **Quality Audit Mechanism:** Continuous clinical data review, AE tracking, and pharmacokinetic blood sample tracking

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14. Observation & Follow-Up Schedule

**Total Duration:** Monitored continuously until disease progression, death, or trial conclusion. **Onsite Visit Cadence:** Concentrated during Cycle 1 (Days 1, 8, 15) for PK, subsequently on Day 1 of subsequent 28-day cycles. **Remote Monitoring Frequency:** Intermittent survival follow-up contacts after treatment discontinuation. **Checklist Review Interval:** Routine efficacy scans every 8-12 weeks depending on disease type.

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15. Sign-off & Approval

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|  | Role                     |  | Name |  | Signature     |  | Date |       |  | :---          |  | :--- |  | :--- |  | :--- |  |
|  | Principal Investigator   |  |      |  | [Insert Name] |  |      | _____ |  | [DD-MMM-YYYY] |  |      |  |      |  |      |  |
|  | Clinical Operations Lead |  |      |  | [Insert Name] |  |      | _____ |  | [DD-MMM-YYYY] |  |      |  |      |  |      |  |
|  | Lead Statistician        |  |      |  | [Insert Name] |  |      | _____ |  | [DD-MMM-YYYY] |  |      |  |      |  |      |  |